Road Traffic Accidents Analysis

Data Analysis Project

Prepared by:

1. Abdullah Hassan Omar
2. Ahmed Fathy Kamal
3. Mohamed Sayed
4. Jana Ahmed Ramzy
5. Basmala Ahmed
6. Jomana Abdelaziz

Supervised by:

Dr. Amal Mahmoud

Table of Contents

Abstract

Road traffic accidents remain one of the leading causes of injuries and fatalities worldwide. This project aims to analyze road traffic accident data to identify patterns, trends, and key risk factors affecting road safety. The analysis was performed using Excel for initial data preparation, MySQL for data storage and querying, Python for data preprocessing and analysis, and Tableau for creating interactive dashboards.

The project explores accident distribution by road type, speed limit, location, vehicle type, driver age, gender, casualty severity, and road user category. The developed dashboards provide clear visualizations that help identify high-risk cities, the most dangerous road types, and the demographic groups most affected by road accidents.

The findings reveal that urban areas experience the majority of collisions, single carriageway roads have the highest accident frequency, and drivers aged 26–35 are the most involved in road accidents. These insights can support authorities and decision-makers in improving road safety strategies, reducing accident rates, and enhancing transportation planning through data-driven decisions.

Chapter 1: Introduction

1.1 Introduction

Road traffic accidents are one of the most significant public safety issues worldwide, causing injuries, fatalities, and economic losses every year. This project analyzes road accident data to identify accident patterns, understand the factors contributing to collisions, and provide data-driven insights that can support road safety improvements.

1.2 Objectives

The main objectives of this project are:

Analyze road traffic accident data.

Identify the most accident-prone road types and locations.

Examine casualty severity levels.

Analyze accidents based on vehicle type, driver age, gender, and road user type.

Study the relationship between speed limits and collision frequency.

Create interactive dashboards for effective data visualization and decision-making.

1.3 Motivation

Road safety is a critical concern that affects millions of people every year. By analyzing accident data, authorities can better understand risk factors and implement preventive measures. This project aims to transform raw accident data into meaningful insights that can contribute to safer roads and reduced accident rates.

1.4 Challenges

During the project, several challenges were encountered:

Handling missing and inconsistent data.

Cleaning large datasets and removing duplicates.

Integrating data from multiple tables.

Ensuring data quality before analysis.

Designing dashboards that effectively communicate insights.

Chapter 2: Methodology

2.1 Dataset Creation

The dataset was obtained from the UK Road Safety Open Data source. Multiple accident-related tables were collected and integrated to create a comprehensive dataset containing information about:

Accident locations

Vehicle types

Driver age groups

Casualty severity levels

Road types

Speed limits

Urban and rural areas

Road user categories

2.2 Tools Used

Excel

- Initial data cleaning and preparation

MySQL

- Data storage, querying, and management

Python

- Data preprocessing, transformation, and analysis

Tableau

- Dashboard creation and visualization

Power BI

- Creating interactive reports and business intelligence visualizations

2.3 Data Cleaning

The following cleaning steps were performed:

Removed duplicate records.

Handled missing values.

Corrected data types.

Standardized column names.

Removed irrelevant or invalid records.

Verified data consistency and accuracy.

2.4 Data Preprocessing

Several preprocessing techniques were applied:

Merged related tables using common keys.

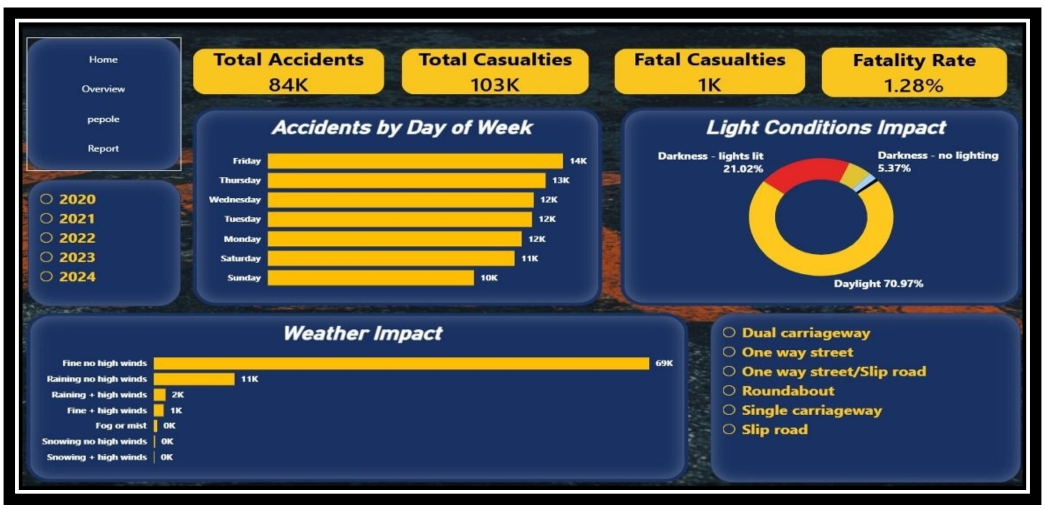
Created age group categories.

Classified casualty severity levels.

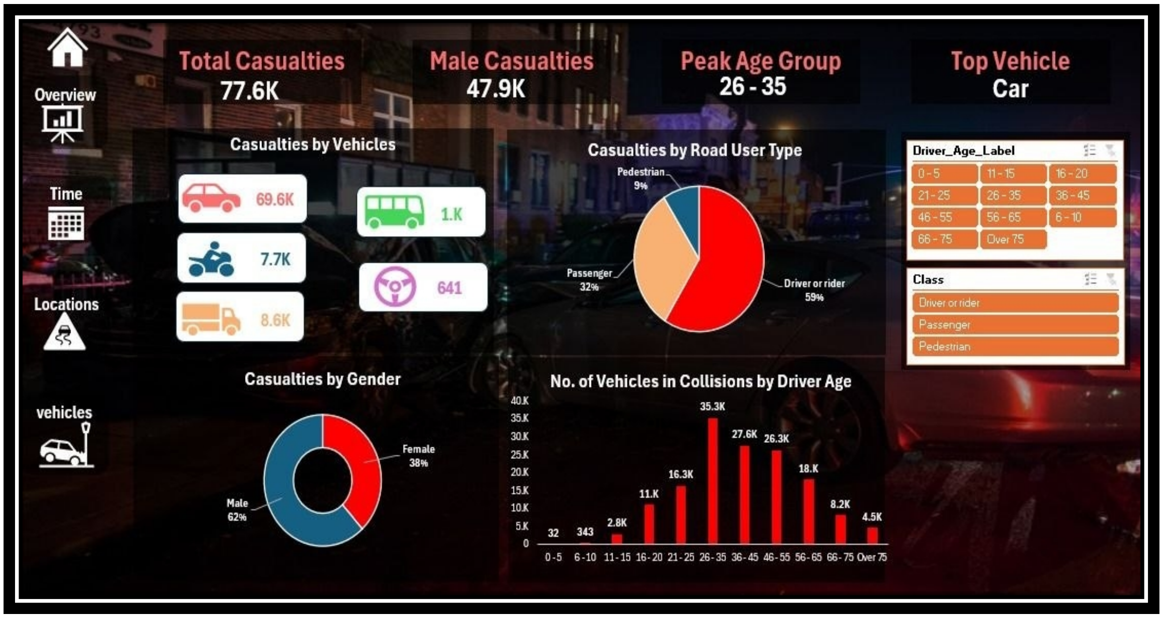
Prepared calculated fields for dashboard metrics.

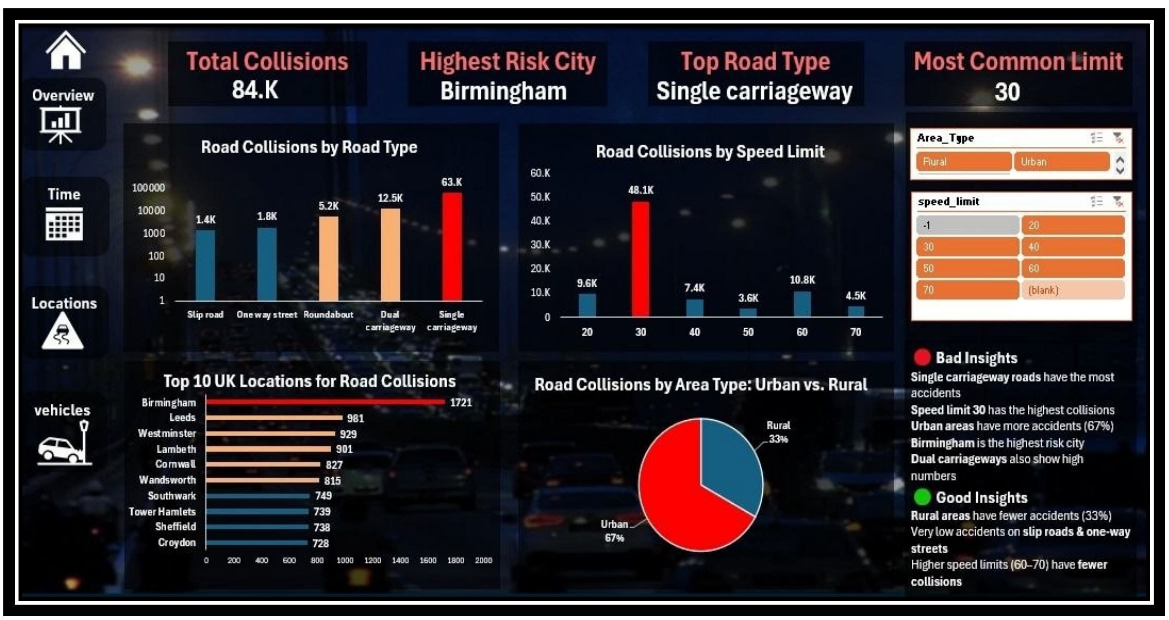
Aggregated data for visual analysis

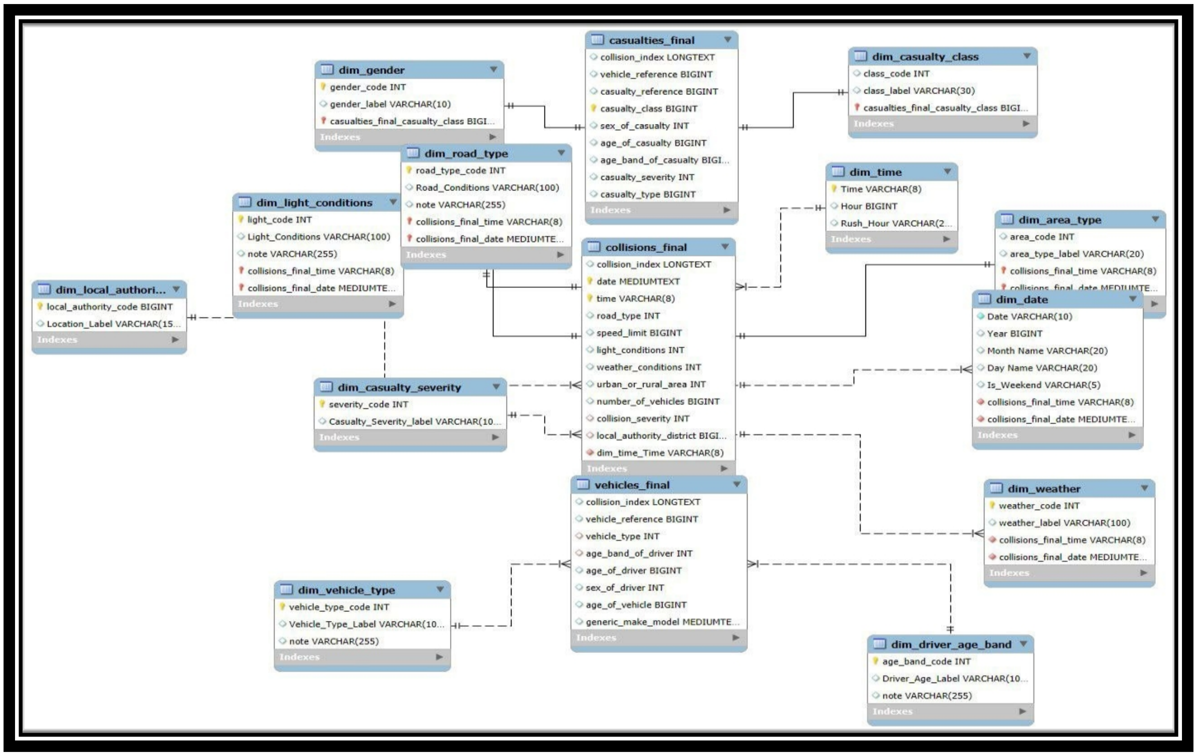
2.5 Screen shoots

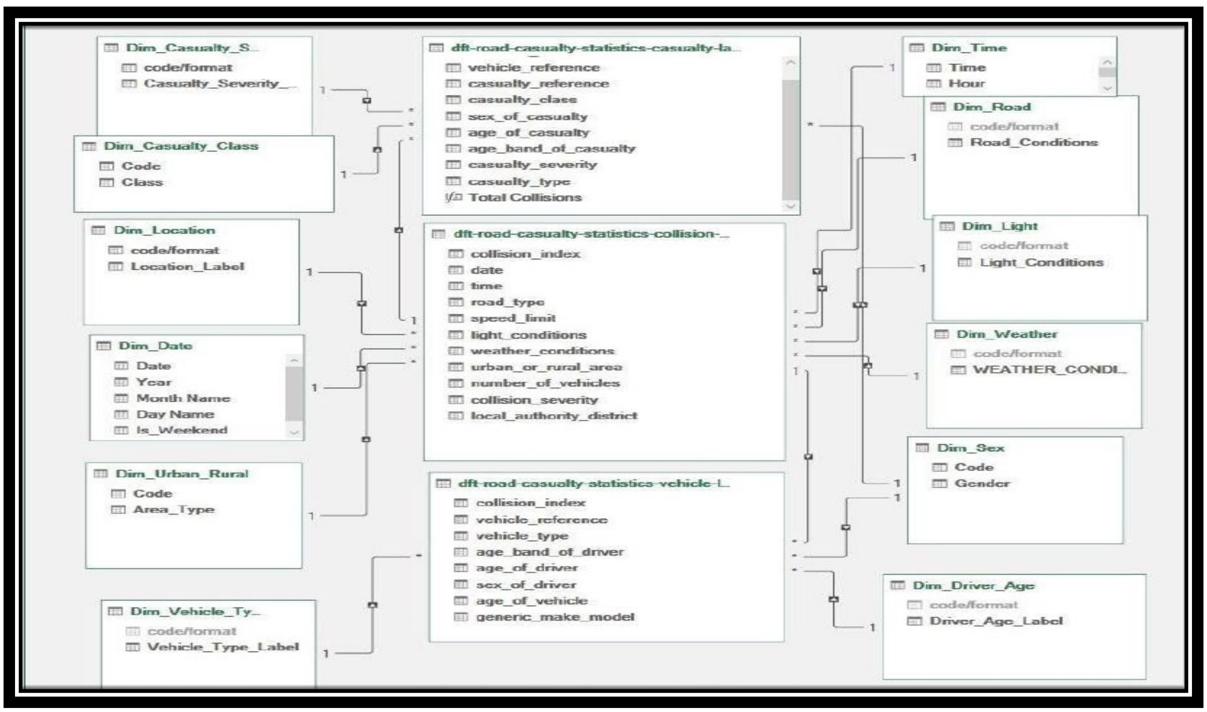


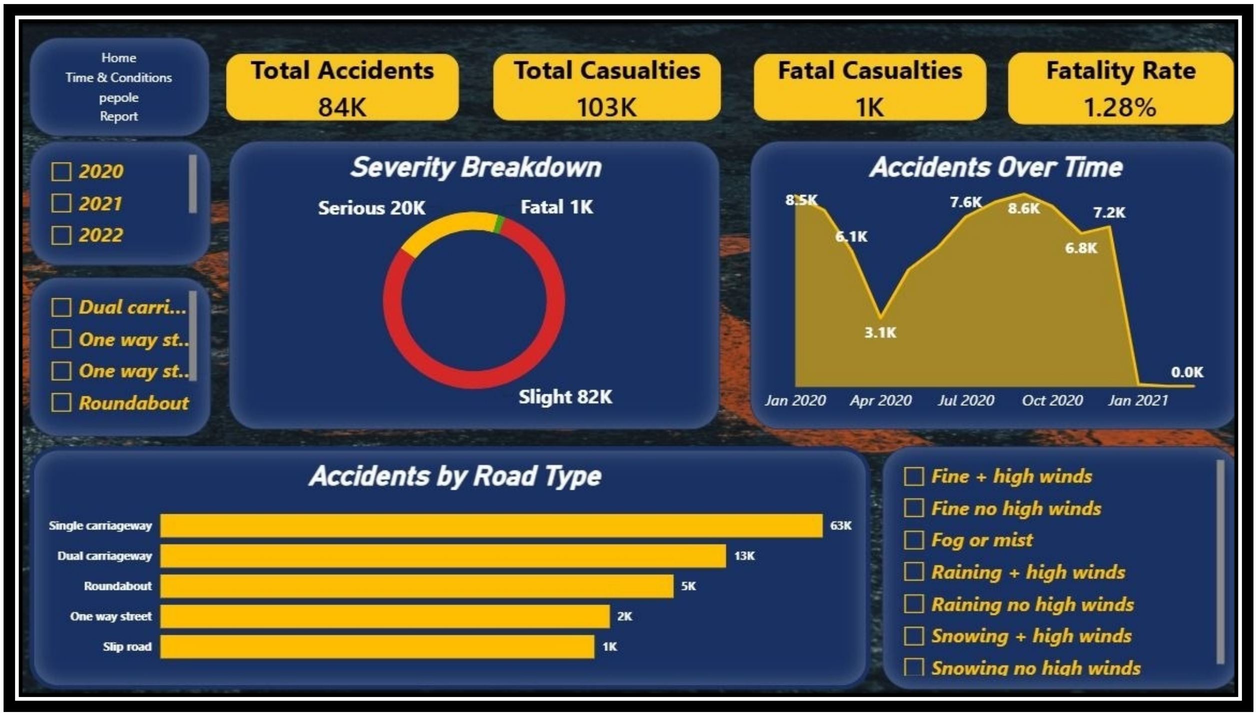
```
Fatal Casualties =
CALCULATE(
    COUNTROWS(casualties_final),
    dim_casualty_severity[Casualty_Severity_label] = "Fatal"
)
```






casualties_final
collision_index LONGTEXT
vehicle_reference BIGINT
casualty_reference BIGINT
casualty_class BIGINT
sex_of_casualty INT
age_of_casualty BIGINT
age_band_of_casualty BIGI...
casualty_severity INT
casualty_type BIGINT
Indexes
dim_gender
gender_code INT
gender_label VARCHAR(10)
casualties_final_casualty_class BIGI...
Indexes
dim_casualty_class
class_code INT
class_label VARCHAR(30)
casualties_final_casualty_class BIGI...
Indexes
dim_road_type
road_type_code INT
Road_Conditions VARCHAR(100)
note VARCHAR(255)
collisions_final_time VARCHAR(8)
collisions_final_date MEDIUMTE...
Indexes
dim_light_conditions
light_code INT
Light_Conditions VARCHAR(100)
note VARCHAR(255)
collisions_final_time VARCHAR(8)
collisions_final_date MEDIUMTE...
Indexes
dim_time
Time VARCHAR(8)
Hour BIGINT
Rush_Hour VARCHAR(2...
Indexes
dim_area_type
area_code INT
area_type_label VARCHAR(20)
collisions_final_time VARCHAR(8)
collisions_final_date MEDIUMTE...
dim_local_authori...
local_authority_code BIGINT
Location_Label VARCHAR(15...
Indexes
collisions_final
collision_index LONGTEXT
date MEDIUMTEXT
time VARCHAR(8)
road_type INT
speed_limit BIGINT
light_conditions INT
weather_conditions INT
urban_or_rural_area INT
number_of_vehicles BIGINT
collision_severity INT
local_authority_district BIGI...
dim_time_Time VARCHAR(8)
Indexes
dim_date
Date VARCHAR(10)
Year BIGINT
Month Name VARCHAR(20)
Day Name VARCHAR(20)
Is_Weekend VARCHAR(5)
collisions_final_time VARCHAR(8)
collisions_final_date MEDIUMTE...
Indexes
dim_casualty_severity
severity_code INT
Casualty_Severity_label VARCHAR(10...
Indexes
vehicles_final
collision_index LONGTEXT
vehicle_reference BIGINT
vehicle_type INT
age_band_of_driver INT
age_of_driver BIGINT
sex_of_driver INT
age_of_vehicle BIGINT
generic_make_model MEDIUMTE...
Indexes
dim_weather
weather_code INT
weather_label VARCHAR(100)
collisions_final_time VARCHAR(8)
collisions_final_date MEDIUMTE...
Indexes
dim_vehicle_type
vehicle_type_code INT
Vehicle_Type_Label VARCHAR(10...
note VARCHAR(255)
Indexes
dim_driver_age_band
age_band_code INT
Driver_Age_Label VARCHAR(10...
note VARCHAR(255)
Indexes


Dim_Casualty_S...
code/format
Casualty_Severity_...
dft-road-casualty-statistics-casualty-la...
vehicle_reference
casualty_reference
casualty_class
sex_of_casualty
age_of_casualty
age_band_of_casualty
casualty_severity
casualty_type
Total Collisions
Dim_Time
Time
Hour
Dim_Road
code/format
Road_Conditions
Dim_Casualty_Class
Code
Class
Dim_Location
code/format
Location_Label
dft-road-casualty-statistics-collision-...
collision_index
date
time
road_type
speed_limit
light_conditions
weather_conditions
urban_or_rural_area
number_of_vehicles
collision_severity
local_authority_district
Dim_Light
code/format
Light_Conditions
Dim_Date
Date
Year
Month Name
Day Name
Is_Weekend
Dim_Weather
code/format
WEATHER_CONDI...
Dim_Urban_Rural
Code
Area_Type
dft-road-casualty-statistics-vehicle-l...
collision_index
vehicle_reference
vehicle_type
age_band_of_driver
age_of_driver
sex_of_driver
age_of_vehicle
generic_make_model
Dim_Sex
Code
Gender
Dim_Vehicle_Ty...
code/format
Vehicle_Type_Label
Dim_Driver_Age
code/format
Driver_Age_Label



5. Feature Engineering

```python
# Extract Features from Date

collisions_df["year"] = collisions_df["date"].dt.year
collisions_df["month"] = collisions_df["date"].dt.month
collisions_df["month_name"] = collisions_df["date"].dt.month_name()

collisions_df["day"] = collisions_df["date"].dt.day
collisions_df["day_name"] = collisions_df["date"].dt.day_name()

print(collisions_df[
    ["date","year","month","month_name","day","day_name"]
].head())
```

[9]

```
...
         date  year  month month_name  day   day_name
0 2021-05-22  2021      5        May   22   Saturday
1 2021-10-20  2021     10    October   20  Wednesday
2 2020-12-01  2020     12   December    1    Tuesday
3 2021-12-09  2021     12   December    9   Thursday
4 2021-04-12  2021      4      April   12     Monday
```

```python
# Extract Hour

collisions_df["hour"] = pd.to_datetime(
    collisions_df["time"].astype(str)
).dt.hour

print(collisions_df[["time","hour"]].head())
```

[10]

```
...
       time  hour
0  22:44:00    22
1  15:50:00    15
2  18:00:00    18
3  16:55:00    16
4  09:02:00     9
```

Road Safety Data Analysis (UK)

Business Problem

Road traffic accidents are a major public safety concern. By analyzing historical road safety data, we can identify accident patterns, high-risk conditions, and factors contributing to accident severity.

Objectives

- Explore road accident trends.
- Analyze accident severity.
- Investigate vehicle and casualty characteristics.
- Identify temporal and environmental patterns.
- Generate insights to support road safety improvements.

Dataset Description

This project uses the UK Road Safety datasets:

- **Collision Dataset:** Information about each accident.
- **Vehicle Dataset:** Information about vehicles involved.
- **Casualty Dataset:** Information about casualties involved.

Each dataset is linked using **accident_index**.

2. Load the data

```python
# Load Datasets

collisions_df = pd.read_csv(
    r"D:\DEPI\Data Analysis\Final Project\Road_Safety_Analysis\data\collision.csv",
    low_memory=False
)

vehicles_df = pd.read_csv(
    r"D:\DEPI\Data Analysis\Final Project\Road_Safety_Analysis\data\vehicle.csv",
    low_memory=False
)

casualties_df = pd.read_csv(
    r"D:\DEPI\Data Analysis\Final Project\Road_Safety_Analysis\data\casualty.csv",
    low_memory=False
)

print("Collisions :", collisions_df.shape)
print("Vehicles   :", vehicles_df.shape)
print("Casualties :", casualties_df.shape)
```

[3]

```
Collisions : (503475, 44)
Vehicles   : (920692, 32)
Casualties : (640522, 23)
```

7. Exploratory Data Analysis (EDA)

7.1 Distribution of Collision Severity

2 cells hidden ...

7.2 Trends Over Time

6 cells hidden ...

7.3 Environmental & Road Conditions

5 cells hidden ...

7.4 Severity vs. Contributing Factors

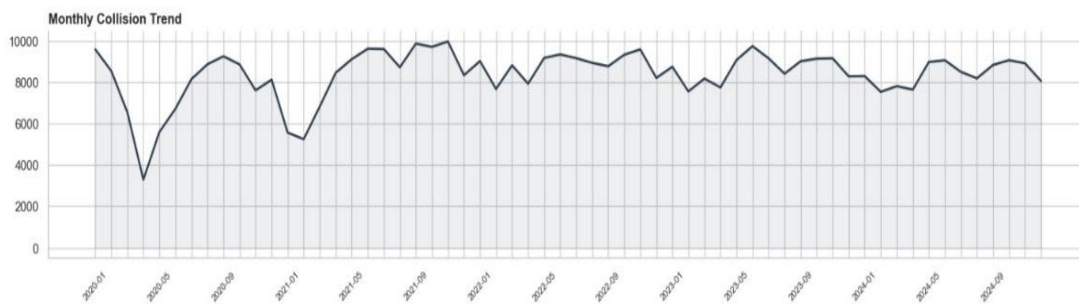
5 cells hidden ...

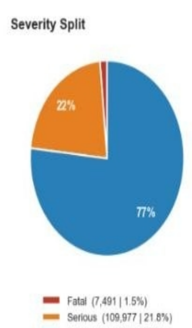
7.5 Casualties & Correlation

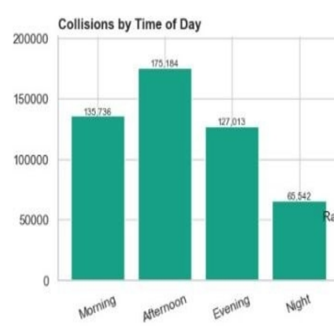
3 cells hidden ...

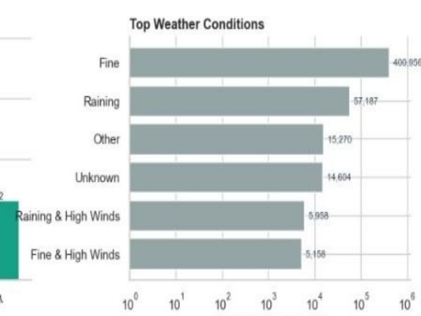
UK Road Safety — Executive Dashboard

503,475	7,491 (1.5%)	109,977
TOTAL COLLISIONS	FATAL	SERIOUS









Chapter 3: Insights & Findings

3.1 Overview Dashboard Insights

Key Findings

Total collisions reached 84.4K accidents.

Total casualties were 77.6K.

Approximately 97.3K vehicles were involved in accidents.

Fatal casualties represented only 1.5K cases.

Cars accounted for the largest number of casualties (69.6K).

Drivers and riders represented the highest casualty group (59%).

Male casualties (62%) exceeded female casualties (38%).

Most recorded casualties were classified as slight injuries, indicating that minor accidents occur more frequently than severe ones.

3.2 Location Dashboard Insights

Key Findings

Birmingham recorded the highest number of road collisions.

Single carriageway roads had the highest accident frequency.

Speed limit 30 mph recorded the largest number of collisions.

Urban areas accounted for approximately 67% of all accidents.

Rural areas represented only 33% of total collisions.

Dual carriageways also showed relatively high accident numbers compared to other road types.

Interpretation

The higher accident rate in urban areas may be attributed to:

Higher traffic density.

Increased vehicle interactions.

More pedestrians and intersections.

Frequent stopping and turning movements.

3.3 Vehicle & Demographic Dashboard Insights

Key Findings

Cars were the most frequently involved vehicle type.

The age group 26–35 years recorded the highest involvement in accidents.

Male casualties significantly exceeded female casualties.

Drivers represented the largest proportion of affected road users.

Pedestrians accounted for the smallest share of casualties.

Interpretation

The 26–35 age group is often the most active driving population, resulting in greater road exposure and a higher likelihood of being involved in collisions.

Recommendations

1. Based on the analysis, the following recommendations are proposed:
2. Increase road safety measures on single carriageway roads.
3. Enhance traffic monitoring in areas with 30 mph speed limits.
4. Implement awareness campaigns targeting drivers aged 26–35 years.
5. Focus accident prevention strategies on high-risk cities such as Birmingham.
6. Improve traffic management and infrastructure in urban areas.
7. Encourage continuous data monitoring to support evidence-based road safety decisions.

Conclusion

This project demonstrates how data analytics tools such as Excel, MySQL, Python, and Tableau can be used to transform raw accident data into actionable insights. The analysis identified key risk factors related to road types, speed limits, vehicle categories, driver demographics, and accident locations. The developed dashboards provide a clear and interactive view of accident trends, supporting decision-makers in improving road safety and reducing future collisions and casualties.